

AI-Powered Resume Screener with Workflow Automation using n8n

**Vocational Training Based Project Report Submitted Under CSVTU
Foundation for Rural Technology & Entrepreneurship , Bhilai**



for

Completion of 4th Semester Vocational Training of the degree

of

BACHELOR OF TECHNOLOGY (HONORS)

In

DATA SCIENCE

(COMPUTER SCIENCE & ENGINEERING)

By

Anil Nai

B.Tech.(Hon) 4th Semester

Roll No. 300012824009

Enrollment No. CE3692

Under the Guidance of

**Mr. Abhinav Claudius & Faculty Supervisor
CEO & Co-Founder, AIMantraas | Assistant Professor**

DECLARATION BY THE CANDIDATE

I, the undersigned, solemnly declare that the vocational-training-based project entitled "AI-POWERED RESUME SCREENER WITH WORKFLOW AUTOMATION USING n8n" is based on my own work carried out during the 4th semester of my graduation under the supervision of **Mr. Abhinav Claudius (CEO & Co-Founder, AIMantraas)**.

____(Signature)_____

Anil Nai

Roll No. **300012824009**

Enrollment No. **CE3692**

CERTIFICATE BY THE SUPERVISOR

This is to certify that the incorporation in the project "AI-Powered Resume Screener with Workflow Automation using n8n" documents the vocational-training-based project work carried out by **Anil Nai**, with Roll No. **300012824009** and Enrollment No. **CE3692**, under the guidance and supervision required for the completion of the 4th-semester vocational training of Bachelor of Technology (Honors) in Artificial Intelligence (Computer Science & Engineering) at Chhattisgarh Swami Vivekananda Technical University, Bhilai (C.G.), India.

To the best of my knowledge and belief the project work:

- Embodies the work of the candidate himself/herself,
- Has duly been completed in the specified time,
- Fulfills the requirement of the Ordinance relating to the B.Tech.(Honors) degree of the University, and
- Is up to the desired standard both in respect of contents and language for being referred to the examiners.

(Signature of H.O.D.)

Dr. J.P. Patra

Associate Professor & HOD

Department of CSE

(Signature of Supervisor)

Mr. Abhinav Claudius

& Assistant Professor

Department of CSE

ACKNOWLEDGEMENT

The real spirit of achieving a goal is through excellence, continuous accountability, and serious discipline. I want to express my deepest gratitude to **INFO CSVTU-FORTE (Foundation for Rural Technology & Entrepreneurship)** and **AIMantraas** for providing me with state-of-the-art software tools, hardware labs, and expert mentorship to complete my Vocational Training on Artificial Intelligence and Internet of Things (AI & IoT).

I express my heartfelt gratitude to **Dr. Pankaj Kumar Mishra**, Director UTD, CSVTU Bhilai, for his constant encouragement and leadership. My profound thanks go to **Dr. J.P. Patra**, Associate Professor & HOD, Department of Computer Science & Engineering, for his invaluable academic guidance.

I am deeply thankful to **Mr. Agranshu Dwivedi** (CEO, INFO CSVTU-FORTE) for building an inspiring startup ecosystem at MSME Tool Room Centre, Durg, that enables students to learn industry-ready skills beyond traditional classroom boundaries.

A special note of thanks to my esteemed mentors:

- Mr. Abhinav Claudius (CEO & Co-Founder, AIMantraas) for his transformative sessions on AI content automation, Transformer architectures, and developer accountability.
- Mr. Rahul Sharwa: for an insightful hands-on masterclass on AI Workflow using n8n.
- Mr. Amit Kumar (Solution Architect at DAZN, ex-IBM, HP, Boeing) whose careerdefining session provided invaluable insights into Model Context Protocol (MCP), AI Security & Governance, AI Agents, and the foundational principle: "The day you stop learning, you are dead." Above all, I render my gratitude to my family and the Almighty for bestowing the strength, ability, and self-confidence to successfully execute this work.

Anil Nai

Roll No. **300012824009**

Enrollment No. **CE3692**

ABSTRACT

This report presents the work carried out during a one-month vocational training program, focused on the design and development of an AI-Powered Resume Screener integrated with workflow automation using n8n. The objective of the project was to build a system capable of automatically evaluating how well a candidate's resume matches a given job description, and to automate the resulting communication and record-keeping process without manual intervention.

The methodology adopted uses Term Frequency–Inverse Document Frequency (TF-IDF) to convert resume and job description text into numerical vectors representing word importance, followed by Cosine Similarity to compute a quantitative match score between each resume and the job description. The scikit-learn library was used to implement this pipeline in Python. Once a match score is generated, an n8n workflow — triggered through a webhook — evaluates the score against a threshold and automatically routes a notification (e.g., shortlisting email or logged entry) accordingly, removing the need for manual follow-up.

During the training, sample resumes and a job description were processed through the pipeline, and the system successfully ranked candidates by relevance, correctly identifying the most closely matching resume with a similarity score of 47.64%, while ranking less relevant resumes lower. The project demonstrates a practical, lightweight approach to resume screening that does not require deep learning infrastructure, making it suitable for small organizations and academic learning alike.

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List of Abbreviations

1. AI – Artificial Intelligence
2. ML – Machine Learning
3. NLP – Natural Language Processing
4. TF-IDF – Term Frequency–Inverse Document Frequency
5. RAG – Retrieval-Augmented Generation
6. API – Application Programming Interface
7. JD – Job Description
8. HR – Human Resources

CHAPTER – I

INTRODUCTION

1.1 Background Information

Recruitment teams routinely receive large volumes of resumes for a single job opening, and manually reading each one against a job description is time-consuming and inconsistent. Natural Language Processing (NLP) offers a way to automatically compare the text of a resume with the text of a job description and produce an objective relevance score. This vocational training project explores a lightweight, explainable approach to this problem using classical NLP techniques rather than heavyweight deep learning models, making it practical to build and understand within a short training period.

1.2 Project Objectives

- To build a system that converts resume and job description text into numerical vectors using TF-IDF.
- To compute a similarity/match score between a resume and a job description using Cosine Similarity.
- To rank multiple candidate resumes by their relevance to a job opening.
- To automate the post-scoring workflow (notification, logging) using the n8n automation platform.
- To understand, at a conceptual level, how Retrieval-Augmented Generation (RAG) could extend this system in the future.

1.3 Significance of the Project

The project is directly relevant to the human resources and recruitment technology (HR-Tech) industry, where automated screening tools are increasingly used to reduce manual effort. It also demonstrates a practical, end-to-end AI workflow — from text processing to a working automation pipeline — which is a valuable skill set for entry-level roles in AI/ML and software development.

1.4 Scope and Limitations

The scope of this project is limited to text-based matching using keyword/vector similarity; it does not perform deep semantic understanding of resumes (such as inferring years of experience or verifying claims). The n8n workflow designed in this training is a

conceptual/demonstration workflow rather than a production HR system connected to a live applicant tracking system. The dataset used consists of a small set of representative resumes rather than a large real-world corpus.

1.5 Overview of the Structure

This report is organised into eight chapters. Chapter II describes the training organisation. Chapter III covers the vocational training activities. Chapter IV explains the methodology (TFIDF and Cosine Similarity). Chapter V describes the implementation, including the development environment and code. Chapter VI presents the results. Chapter VII concludes the report and outlines future scope. Chapter VIII reflects on the learning outcomes of the training.

CHAPTER – II

ORGANIZATION OVERVIEW

2.1 Company Profile

CSVТУ Foundation for Rural Technology and Entrepreneurship (CSVТУ-FORTE) was established as a Section-8 company under the guidance of Chhattisgarh Swami Vivekanand Technical University (CSVТУ). The organisation focuses on promoting innovation, entrepreneurship, and sustainable development, particularly in rural and underserved areas.

CSVТУ-FORTE works to create an entrepreneurial ecosystem that supports early-stage startups and grassroots-level innovations. Through initiatives such as the Rural Incubation Centre (RIC) and Rural Entrepreneurship Development Centres (REDCs), the organisation provides aspiring entrepreneurs with mentorship, financial assistance, technical resources, and opportunities to transform innovative ideas into marketable solutions.

2.2 Vision and Mission

Vision

The vision of CSVТУ-FORTE is to build a sustainable ecosystem for innovation and entrepreneurship that transforms grassroots ideas into impactful solutions, particularly for rural and underserved communities. **Mission**

The mission of CSVТУ-FORTE is to promote innovation and entrepreneurship by supporting grassroots-level startups, providing incubation facilities, mentorship, financial assistance, and technical resources, and encouraging sustainable rural development.

2.3 Organizational Structure

CSVТУ-FORTE operates as a Section-8 organisation under the guidance of Chhattisgarh Swami Vivekanand Technical University (CSVТУ). Its activities are organised around promoting innovation, entrepreneurship, incubation, and rural technology development.

2.4 Product and Services

CSVТУ-FORTE primarily provides incubation, entrepreneurship development, innovation support, and skill-development programs rather than conventional commercial products. Its major programs and services include:

- Rural Incubation Centre (RIC): Provides incubation support, mentorship, resources, and guidance to early-stage startups working on rural and grassroots-level solutions.

- Rural Entrepreneurship Development Centres (REDCs): Promote entrepreneurship and innovation among students, aspiring entrepreneurs, and rural communities.
- Ignition Grant under I-TBI: Provides financial support to help entrepreneurs develop and validate innovative ideas.
- MSME Incubation Scheme: Supports the development and commercialization of innovative products and business ideas.

CHAPTER – III

VOCATIONAL TRAINING ACTIVITIES

3.1 Description of Activities

The one-month training covered the fundamentals of Python for data processing, an introduction to core machine learning concepts (text vectorization and similarity measures), and hands-on use of the n8n workflow automation platform. Activities included guided exercises, followed by independent development of the resume screener project as a capstone task.

3.2 Skills Developed

- Python programming for text processing (string handling, loops, functions).
- Practical use of the scikit-learn library — TfidfVectorizer and cosine_similarity.
- Understanding of core NLP concepts: tokenization, stop-word removal, vectorization.
- Building and configuring automation workflows in n8n (Webhook trigger, IF/condition nodes, notification nodes).
- Structuring and documenting a technical project report.

3.3 Challenges Faced

Coming in without a prior machine learning background, the main challenge was grasping vectorization and similarity scoring conceptually before writing any code. This was addressed by focusing on classical, easily explainable techniques (TF-IDF and Cosine Similarity) instead of deep learning, which made the concepts approachable within the training timeframe.

3.4 Achievements and Contributions

Successfully implemented a working resume-matching script that produces a ranked list of candidates from raw text input, and designed an accompanying n8n automation workflow to act on the generated scores.

3.5 Learning Outcomes

The training provided a foundational understanding of how real-world AI/ML systems are structured — from raw data, to a scoring model, to an automated action — and demonstrated that useful, deployable AI tools can be built without requiring large datasets or deep learning infrastructure.

CHAPTER – IV

METHODOLOGY

4.1 Project Overview

The project builds a pipeline that takes a job description and one or more resumes as plain text, converts them into numerical vectors, and computes a similarity score between each resume and the job description. Resumes are then ranked by this score, and the outcome is passed to an n8n workflow for automated follow-up action.

4.2 Research Design or Approach

A quantitative, experimental approach was used: text documents were transformed into vector representations and compared mathematically, rather than through manual/qualitative review. The two core techniques used are described below.

TF-IDF (Term Frequency–Inverse Document Frequency)

TF-IDF converts each document (a resume or a job description) into a vector of numbers, one per word, where the value reflects how important that word is to the document. A word that appears frequently in one resume but rarely across the other documents gets a high weight, while common words (e.g., 'the', 'and') are down-weighted automatically.

Cosine Similarity

Once both the job description and a resume are represented as TF-IDF vectors, Cosine Similarity measures how close the two vectors are by calculating the cosine of the angle between them. The result is a score between 0 and 1 (shown here as a percentage) — a higher score means the resume's wording is more aligned with the job description's wording.

Fig 4.1: TF-IDF and Cosine Similarity Pipeline — Text → TF-IDF Vectors → Cosine Similarity → Match Score

4.3 Data Collection Methods

For this training exercise, one job description and four representative candidate resumes were authored to reflect realistic variation in relevant skills (e.g., a strong technical match, a partial match, and an unrelated profile). In a production setting, this text would instead be extracted from uploaded PDF/DOCX resumes using a parsing library such as pdfplumber or PyPDF2.

4.4 Data Analysis Techniques

The scikit-learn library's `TfidfVectorizer` was used to fit a vocabulary across all documents and transform them into TF-IDF vectors, with English stop-words removed. The `cosine_similarity` function was then applied between the job description vector and each resume vector, and results were ranked in descending order of score using `pandas`.

4.5 Ethical Considerations

Automated resume screening carries a risk of unfairly disadvantaging qualified candidates whose resumes use different wording than the job description, or of encoding bias if trained on biased historical data. Because this system relies on transparent keyword/TF-IDF matching rather than a black-box model, its scoring logic is explainable and auditable. It is recommended that such a tool be used to assist, not replace, human judgement in real recruitment decisions.

4.6 Limitations

TF-IDF matching is based on word overlap and does not capture semantic meaning — a resume that describes the same skill using different terminology (e.g., 'built REST APIs' vs 'developed web services') may score lower than its true relevance. This limitation, and how it could be addressed using semantic embeddings, is discussed further in Chapter VII.

CHAPTER – V

IMPLEMENTATION

5.1 Development Environment

Software Tools

- Python 3 — core programming language
- scikit-learn — TfidfVectorizer, cosine_similarity
- pandas — result ranking and tabulation
- n8n — workflow automation (Webhook, IF, Email/Sheets nodes)
- Flask (proposed extension) — web interface for resume upload

Hardware

A standard laptop (any modern CPU, 8 GB RAM) is sufficient, as TF-IDF and cosine similarity are computationally lightweight and require no GPU.

Configuration

Python 3.x with pip-installed scikit-learn and pandas; n8n run locally via Docker or accessed through n8n's cloud offering.

5.2 Project Execution

The job description and sample resumes were written as text. A single script combined them into one document list, fit a shared TF-IDF vocabulary across all documents, computed the cosine similarity of each resume against the job description, and printed a ranked results table.

The complete script is shown below.

```
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.metrics.pairwise import cosine_similarity import
pandas as pd

documents = [job_description] + list(resumes.values())
vectorizer = TfidfVectorizer(stop_words='english') tfidf_matrix
= vectorizer.fit_transform(documents)

jd_vector = tfidf_matrix[0:1]
resume_vectors = tfidf_matrix[1:]
scores = cosine_similarity(jd_vector, resume_vectors)[0]

results = pd.DataFrame({
    'Resume': list(resumes.keys()),
    'Match Score (%)': [round(s * 100, 2) for s in scores]
}).sort_values(by='Match Score (%)', ascending=False)
```

Fig 5.1: Core resume-matching script using TF-IDF and Cosine Similarity

5.3 Timeline

Week	Activity
Week 1	Python refresher; introduction to NLP concepts (tokenization, TF-IDF)
Week 2	Hands-on with scikit-learn; built the TF-IDF + cosine similarity matcher
Week 3	Introduction to n8n; designed webhook-triggered notification workflow
Week 4	Integration, testing, and report documentation

5.4 Resource Allocation

The project required no paid resources — all software used (Python, scikit-learn, pandas, n8n's free/self-hosted tier) is free and open-source, making it a low-cost, reproducible solution suitable for small teams or academic settings.

5.5 Challenges Faced

Designing an n8n workflow that correctly branches on the match-score threshold required understanding n8n's IF node and expression syntax, which was new. This was resolved by testing the workflow with dummy webhook payloads before wiring it to the actual scoring script.

5.6 Success Factors

Keeping the ML component intentionally simple (TF-IDF rather than deep learning) allowed more time to be spent understanding the full pipeline end-to-end, including the automation layer, rather than being consumed by model complexity.

5.7 Lessons Learned

A simple, explainable model that is fully understood and correctly integrated into a workflow is often more valuable — especially in a one-month training — than a complex model that is only partially understood.

CHAPTER – VI

RESULTS AND DISCUSSION

6.1 Presentation of Results

The script was run against one job description (seeking a Python/ML developer with Flask, SQL, and Git experience) and four sample resumes. The ranked output is shown in Table 6.1.

Rank	Resume	Match Score (%)
1	Candidate_A_Aditya.txt	47.64
2	Candidate_D_Simran.txt	22.68
3	Candidate_C_Karan.txt	13.95
4	Candidate_B_Riya.txt	7.69

Table 6.1: Resume Match Scores ranked by relevance to the job description

6.2 Interpretation of Results

The candidate with the strongest overlap in relevant technical terms (Python, scikit-learn, pandas, Flask, Git, SQL) scored highest at 47.64%, while the graphic designer's resume — sharing almost no vocabulary with the job description — scored lowest at 7.69%. The two intermediate resumes were correctly ordered based on their partial skill overlap, confirming that the model is sensitive to relevant keyword density.

6.3 Comparison with Objectives

All stated objectives (Section 1.2) were met: a working TF-IDF/cosine-similarity matcher was built, resumes were successfully ranked, and an n8n automation workflow was designed around the scoring threshold.

6.4 Discussion of Key Findings

The results confirm that even a lightweight, classical NLP technique can produce a sensible and explainable ranking for resume screening without requiring a trained neural network or

large dataset. This makes the approach attractive for smaller organisations that need a quick, low-cost screening aid.

6.5 Limitations and Future Directions

Because scores are based purely on shared vocabulary, a resume using synonyms or different phrasing for the same skill could be under-scored. This is discussed further as future scope in Chapter VII, particularly the potential move to semantic embeddings or a RAG-based approach.

CHAPTER – VII

CONCLUSION AND FUTURE SCOPE

7.1 Summary of Findings

This vocational training successfully delivered a working AI-powered resume screening pipeline using TF-IDF and Cosine Similarity, paired with a conceptual n8n automation workflow for downstream notification and logging.

7.2 Achievement of Objectives

All project objectives outlined in Chapter I were achieved within the one-month training period, given the deliberately scoped, classical-NLP approach chosen for the project.

7.3 Implications and Recommendations

The project shows that organisations without dedicated data science teams can still adopt a first layer of AI-assisted screening using simple, transparent techniques, before investing in more complex, embedding-based systems.

7.4 Future Scope

A natural extension of this project is to replace TF-IDF with sentence embeddings (e.g., using the sentence-transformers library) for more semantically aware matching. A further extension is to build a Retrieval-Augmented Generation (RAG) system: resumes and job descriptions would be split into chunks, converted into embeddings, and stored in a vector database (such as FAISS or Chroma). When a query — such as 'summarise this candidate's relevant experience' — is asked, the most relevant chunks are retrieved and passed to a language model, which generates an answer grounded in the actual resume content rather than relying purely on similarity scores. This would allow the screener to answer open-ended recruiter questions, not just produce a numeric score.

7.5 Personal Reflections

Starting this training with no prior machine learning background, the most valuable takeaway was learning to scope a project appropriately — choosing a technique (TF-IDF) that could be

fully understood and correctly implemented within the available time, rather than attempting a more advanced but poorly understood model. This experience has built confidence to explore more advanced NLP techniques, including embeddings and RAG, in future coursework and projects.

CHAPTER – VIII

LEARNING OUTCOMES

8.1 Skills Developed

- Python-based text processing and use of the scikit-learn ML library.
- Workflow automation design using n8n (webhooks, conditional routing).
- Technical documentation and report writing.

8.2 Knowledge Gained

Conceptual understanding of TF-IDF, cosine similarity, and how classical NLP techniques underpin many real-world text-matching systems, as well as an introduction to RetrievalAugmented Generation as a modern extension of these ideas.

8.3 Professional Development

The training highlighted an interest in the intersection of AI/ML and practical automation tooling, and demonstrated the value of combining a modest ML model with a well-designed automation workflow to produce a genuinely usable tool.

8.4 Personal Growth

Working under a tight timeline required prioritising a scoped, achievable solution over an ambitious but incomplete one — a discipline that will be useful in future academic and professional projects.

8.5 Future Application

The skills gained will be applied to future academic projects (including GATE CS preparation topics touching on algorithms and data structures) and to personal web-development projects, by incorporating lightweight ML features into applications already being built.

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